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RESTful Bookstore API

Introduction

Modern applications rely heavily on backend services that communicate using REST APIs. These APIs allow different systems to exchange data efficiently over HTTP protocols. The objective of this project is to design and implement a RESTful API that manages bookstore data.

The application enables users to perform CRUD operations on books while maintaining structured data storage using a relational database. The project also demonstrates the use of Spring Boot for building scalable backend applications with minimal configuration.

Abstract

The RESTful Bookstore API is a backend application developed using Spring Boot that provides an interface for managing books in a bookstore system. The API allows users to perform operations such as adding new books, retrieving book details, deleting books, and filtering books by author. It uses Spring Data JPA for database interaction and H2 as an in-memory database for storage. The API is tested using Postman, and Swagger UI is integrated to provide interactive API documentation. This project demonstrates the implementation of RESTful services, database connectivity, and backend application architecture using Java.

Tools and Technologies Used

Tool / Technology	Purpose
Java	Core programming language
Spring Boot	Framework for building REST APIs
Spring Data JPA	Database interaction and ORM
H2 Database	In-memory database for testing
Postman	API testing
Swagger UI	API documentation and testing

Tool / Technology	Purpose
Maven	Dependency management and build tool
Eclipse IDE	Development environment

Steps Involved in Building the Project

1. Created a Spring Boot project using Spring Initializr with required dependencies.
2. Designed the **Book entity class** representing book attributes such as title, author, and price.
3. Implemented the **BookRepository interface** using Spring Data JPA for database operations.
4. Developed the **BookController class** to expose REST API endpoints for CRUD operations.
5. Implemented APIs for retrieving books, adding new books, deleting books, and filtering books by author.
6. Added **pagination and sorting features** to improve data retrieval performance.
7. Tested API endpoints using **Postman** to ensure proper request-response functionality.
8. Integrated **Swagger UI** to automatically generate API documentation for easy testing and demonstration.

Conclusion

The RESTful Bookstore API project successfully demonstrates the development of a backend application using Spring Boot. The system provides efficient management of bookstore data through RESTful endpoints and supports operations such as creating, retrieving, filtering, and deleting records. The integration of Swagger UI and Postman testing ensures that the API is well-documented and easy to use. This project enhanced understanding of REST architecture, backend development, and database interaction using Java and Spring Boot.